

HubSpot Sync - Contact Management Integration

A full-stack application that integrates with HubSpot, automatically synchronizes contacts, and keeps notes synchronized between the application and HubSpot.

Live Demo

Component	URL
Frontend	https://hubspot-sync-frontend.vercel.app
Backend	https://hubspot-sync-backend.vercel.app
GitHub	https://github.com/Yashvar2004/assignment

Table of Contents

- [How to Use the Deployed App](#)
 - [Project Overview](#)
 - [Architecture](#)
 - [Technology Choices](#)
 - [Features Implemented](#)
 - [Setup Instructions](#)
 - [Environment Variables](#)
 - [API Documentation](#)
 - [Scalability Design](#)
 - [Limitations](#)
 - [Future Improvements](#)
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How to Use the Deployed App

Step 1: Get Your HubSpot PAT (Personal Access Token)

1. Go to <https://app.hubspot.com>
2. Click **Settings** (gear icon, top right corner)
3. In the left sidebar, go to **Integrations** → **Private Apps**
4. If you don't see "Private Apps", search for "**Legacy Apps**" in the settings search bar
5. Click "**Create a private app**"
6. Give it a name: `My HubSpot Sync App`
7. Go to the **Scopes** tab and add these scopes:
 - `crm.objects.contacts.read`
 - `crm.objects.contacts.write`
 - `crm.objects.companies.read`
 - `crm.objects.deals.read`
 - `crm.objects.deals.write`
8. Click "**Create app**"
9. Copy the **Access Token** (starts with `pat-na2-...`)

Step 2: Sign Up on the Deployed App

1. Open <https://hubspot-sync-frontend.vercel.app>

- 2. Click **"Don't have an account? Sign up"**
- 3. Enter your **email** and **password** (minimum 6 characters)
- 4. Click **"Create Account"**
- 5. You will see the **"Connect HubSpot"** page
- 6. Paste your **PAT token** (from Step 1) in the input field
- 7. Click **"Connect HubSpot"**
- 8. Wait **10-15 seconds** — your contacts will sync automatically

Step 3: Use the App

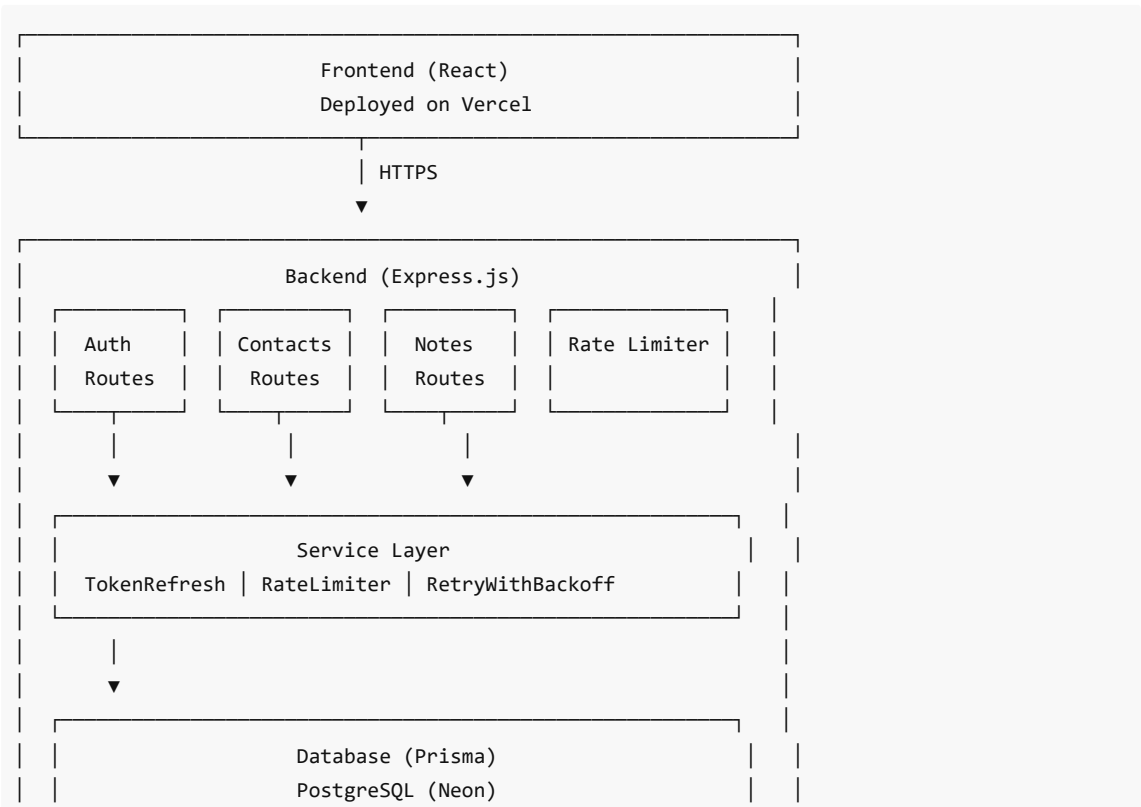
- **View contacts:** Your HubSpot contacts appear automatically
- **Add notes:** Click on any contact, type a note, and click "Add Note"
- **Notes sync to HubSpot:** Notes you add in the app appear in HubSpot automatically
- **Auto-sync:** New contacts from HubSpot sync every 30 seconds
- **Refresh:** Click the "Refresh" button for instant sync

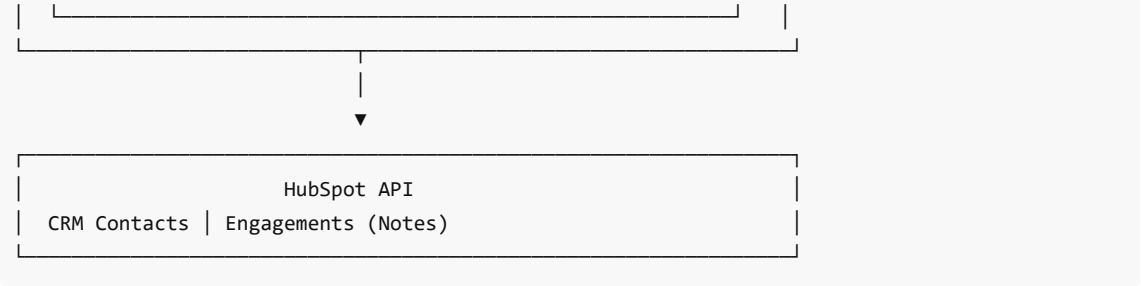
Project Overview

This application provides:

- **User Authentication** — Register and login with email/password
- **HubSpot Integration** — Connect your HubSpot account using a PAT token
- **Automatic Contact Synchronization** — Contacts sync automatically after connection
- **Bidirectional Note Sync** — Notes created in the app sync to HubSpot
- **Auto-refresh** — Contacts sync every 30 seconds automatically
- **Multi-user Support** — Each user has their own contacts and data

Architecture





Technology Choices

Backend

Technology	Purpose
Node.js 22	Runtime environment
Express.js	Web framework
Prisma	ORM for database
PostgreSQL (Neon)	Database
express-rate-limit	API rate limiting
jsonwebtoken	JWT authentication
axios	HTTP client with retry

Frontend

Technology	Purpose
React 18	UI framework
TypeScript	Type safety
Vite	Build tool
Tailwind CSS	Styling
React Router	Client-side routing

Features Implemented

Feature	Status
User registration/login	✓
HubSpot connection via PAT token	✓
Automatic contact sync	✓
Contact list with search	✓

Contact details view	✓
Add notes to contacts	✓
Notes sync to HubSpot	✓
Auto-refresh every 30 seconds	✓
Rate limiting	✓
Retry with exponential backoff	✓
Cursor-based pagination	✓
Idempotent operations (upsert)	✓
Token refresh	✓
Error handling	✓
Loading states	✓

Setup Instructions

Prerequisites

- Node.js 18+
- PostgreSQL database (Neon recommended)
- HubSpot account with Private App

1. Clone the Repository

```
git clone https://github.com/Yashvar2004/assignment.git
cd assignment
```

2. Backend Setup

```
cd backend
npm install
```

Create `.env` file:

```
DATABASE_URL="postgresql://user:password@host:5432/dbname?sslmode=require"
JWT_SECRET="your-secure-random-string"
HUBSPOT_PAT_TOKEN="pat-na2-your-token"
FRONTEND_URL="http://localhost:5173"
NODE_ENV="development"
```

```
npx prisma generate
npx prisma db push
```

```
npm run dev
```

3. Frontend Setup

```
cd frontend
npm install
npm run dev
```

4. Open Application

Navigate to `http://localhost:5173`

Environment Variables

Backend

Variable	Required	Description
DATABASE_URL	Yes	PostgreSQL connection string
JWT_SECRET	Yes	Secret for JWT signing (min 32 characters)
HUBSPOT_PAT_TOKEN	No	HubSpot Personal Access Token
FRONTEND_URL	No	Frontend URL for CORS
NODE_ENV	No	development OR production

API Documentation

Authentication

Method	Endpoint	Description
POST	<code>/api/auth/register</code>	Register new user
POST	<code>/api/auth/login</code>	Login
GET	<code>/api/auth/me</code>	Get current user
POST	<code>/api/auth/connect-hubspot</code>	Connect HubSpot with PAT token
GET	<code>/api/auth/status</code>	Check connection status

Contacts

Method	Endpoint	Description
GET	<code>/api/contacts</code>	List contacts (paginated, searchable)
GET	<code>/api/contacts/:id</code>	Get contact details
GET	<code>/api/contacts/sync/jobs</code>	Get sync jobs

Notes

Method	Endpoint	Description
POST	/api/contacts/:contactId/notes	Create note
GET	/api/contacts/:contactId/notes	Get notes
DELETE	/api/notes/:noteId	Delete note
GET	/api/notes/sync-status	Note sync status

Health

Method	Endpoint	Description
GET	/health	Health check

Scalability Design

Feature	Implementation
Rate Limiting	express-rate-limit (100 req/15min) + HubSpot token bucket (100 req/10s)
Retry	Exponential backoff with jitter (5 attempts)
Idempotent Operations	Upsert with compound unique constraint
Cursor-based Pagination	Resumable syncs with cursor tracking
Batch Processing	Promise.all for parallel upserts
Token Refresh	Auto-refresh 5 minutes before expiry

Limitations

1. **PAT Token Authentication:** Uses PAT token instead of full OAuth flow (HubSpot disabled public app creation)
2. **In-memory Rate Limiting:** Rate limits reset on serverless cold start
3. **No Queue System:** Background jobs run in-process (documented as future improvement)
4. **Single HubSpot Account per User:** Each user connects one HubSpot account

Future Improvements

1. **BullMQ + Redis:** Extract background jobs to dedicated workers
2. **Full OAuth Flow:** Implement when HubSpot re-enables public app creation
3. **Webhook Support:** Real-time updates from HubSpot
4. **Incremental Sync:** Only sync changed contacts
5. **Docker Support:** Containerized deployment
6. **Unit Tests:** Comprehensive test coverage

License

This project is created for technical assessment purposes.